Birzeit University

Faculty of Engineering and Technology

Department of Computer Science

Database Management Systems (COMP333)

Phase 2

Electronics Store Information Management System

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Introduction:

The client wants to implement a database system for their electronics store, which sells a variety of products such as laptops, printers, mice, keyboards, and other accessories. The system will help manage inventory, track sold products, and organize supplier (agent) information efficiently.

This system will help the client deal with unorganized data and help keep track of supplier details, inventory, trends and more. The database will resolve these issues by keeping a clear record of all incoming and outgoing products, ensuring accurate stock management. This system will also provide reports to help the client make informed business decisions, reduce losses, and improve overall store efficiency.

Client information:

The client is the manager of a local electronics store called **الرحال**, which operates under the business number **(No. 2420392)**. With a focus on providing high-quality electronic products such as laptops, printers, mice, and keyboards, the client oversees all aspects of store management, including inventory control, supplier relations, and sales tracking.

Technology:

Name	Processor	RAM	Graphic card	Storage	OS
Jamil Habash	Intel core I5 12400F	16 GB DDR4	GeForce RTX 3050	512GB SSD	Windows 10
Abdallah Kokash	Intel core I7 11370H	16 GB DDR4	GeForce RTX 3060	1T SSD	Windows 10

In this project, we are utilizing a combination of technologies to develop an efficient database-driven application. Python, along with Flask, serves as the backend framework, providing a lightweight yet powerful environment for handling server-side logic and database interactions. We are using MySQL as the database management system to store, manage, and retrieve data efficiently. PyCharm is our primary development environment, offering advanced coding and debugging features. For the frontend, we are leveraging HTML and CSS to create a responsive and user-friendly interface.

Project details:

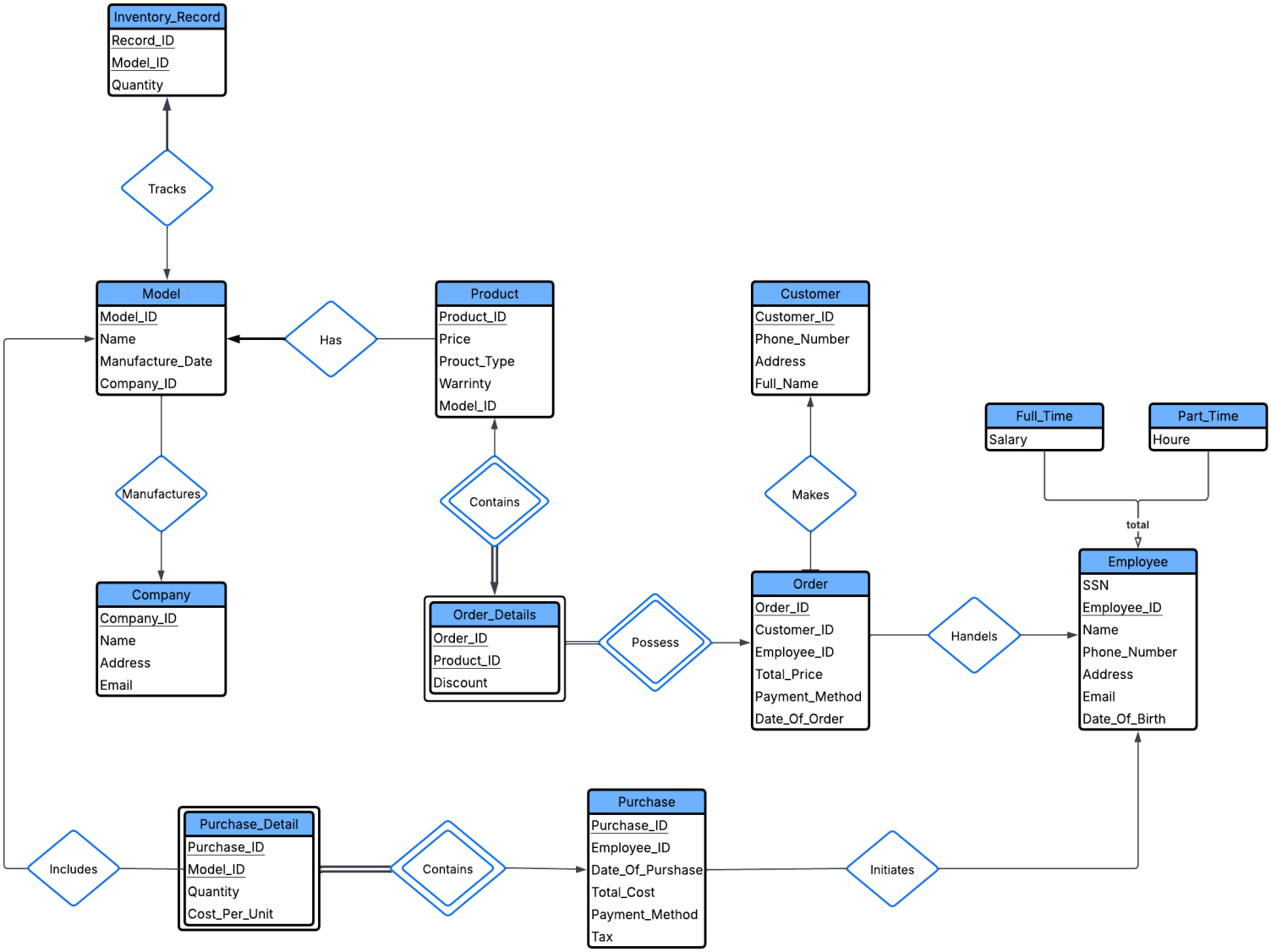
Entities:

- **Products** have an ID, Price, Product Type, Warranty, and Model ID.
- **Customers** have an ID, Phone Number, Address, and Name.
- **Orders** have an ID, Customer ID, Employee ID, Total Price, Payment Method, and purchase date.
- **Employees** have an ID, Name, Social Security Number (SSN), Phone Number, Address, Email, and Birthdate.
- **Employees** can either be part time with price per hour or full time with a salary
- **Order Details** contain an Order ID, Product ID and discount.
- **Companies** have an ID, Name, Address, and Email.
- **Models** have an ID, Name and Company ID, and manufactural date.
- **Purchase** has an ID, Employee ID, Purchase date, Total cost, Payment method, and Tax.
- **Purchase Detail** has ID, Model ID, Quantity, and Cost per unit.
- **Inventory Record** has ID, Model ID, and Quantity.

Relationships:

- A **Customer** can place multiple **Orders**, but each **Order** belongs to only one **Customer**.
- An **Employee** can handle multiple **Orders**, but each **Order** is processed by only one **Employee**.
- An **Employee** can be either **Full-time** or **Part-time**, but not both.
- Each **Product** belongs to a specific **Model**, but a **Model** can have multiple **Products**.
- Each **Model** is created by one **Company**, but a **Company** can manufacture multiple **Models**.
- An **Order** can have multiple **Details** and a **Detail** is tight to an **Order**.
- A **product** has one **Detail** and a **Detail** belongs to one **Product**.
- **A Purchase** contains multiple **Purchase Details**, and each **Purchase Detail** is tied to one **Purchase**.
- Each **Purchase Detail** refers to exactly one **Model**, but a **Model** can appear in multiple **Purchase Details**.
- Each **Purchase** is initiated by one **Employee**, but an **Employee** can initiate multiple **Purchases**.
- An **Inventory Record** tracks the quantity of a specific **Model**, and a **Model** can have one **Inventory Record.**

Queries:

- Retrieve all customer details.
- Retrieve the total number of orders placed by each customer.
- Retrieve the details of all orders handled by a specific employee.
- Retrieve the total sales amount handled by each employee.
- Retrieve all products and their prices.
- Retrieve the details of all orders that contain a specific product.
- Retrieve the total revenue generated from each product.
- Retrieve details of orders placed within a specific date range.
- Retrieve all orders that include a discount.
- Retrieve the model details of all available products.
- Retrieve the customers who have placed the most orders.
- Retrieve the payment methods used for each order.
- Retrieve the most uses payment methods.
- Retrieve the details of the employee who handled the most orders.
- Retrieve the most frequently ordered product.
- Retrieve the total number of orders processed by each employee.
- Retrieve the details of the company that manufactures a specific model
- Retrieve the product that was bought most by a specific customer.
- Retrieve all products that are under warranty
- Retrieve all products that belong to a specific model.
- Retrieve the most expensive order.
- Retrieve the average order value for all customers.
- Retrieve the latest products model added to the database.
- Retrieve the details of all orders placed by a specific customer.
- Retrieve the details of all orders Handled by a specific employee.
- Retrieve the total amount of sales in a specific time range.
- Retrieve the total amount of orders in a specific time range.

Entity-Relationship diagram: